



MODULE NAME:	MODULE CODE:
COMPUTER NETWORKS 1A	CONE5111/p/w
NETWORK ENGINEERING 1A	NWEG5111

ASSESSMENT TYPE: ASSIGNMENT 2 (PAPER AND MARKING RUBRICS)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Save a copy of your assignment before submitting it.***
3. ***Assignments must be typed unless otherwise specified.***
4. ***All work must be adequately and correctly referenced.***
5. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where both minor and major errors** (e.g., two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g., alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g., publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Assessments: Assignment 1 and Assignment 2 — Background and Requirements

In this module, you will be expected to complete two (2) assignments and an unseen written examination. Ensure that you keep a soft copy of the completed assignments. Use your creativity and be as innovative as possible in the creation of the final solution. Use the marking guidelines/ marking rubrics provided for each activity to guide you through the minimum required content that you are required to submit for evaluation. This assignment will require you to work consistently during the semester in order to achieve the best results.

Please read the hardware lab requirements and software lab requirements on page XIX in the prescribed material.

It is important to familiarise yourself with Cisco's Packet Tracer as well as to be able to work with activities and parts of the assessments.

Module Final Mark and Allocation of Marks

Description	Mark Allocation	Weight
Assignment 1	100 marks	25%
Assignment 2	100 marks	30%
ICE tasks		10%
Written Unseen Examination	100 marks	35%
TOTAL		100%

BACKGROUND

PLEASE NOTE:

*Assignment 2 will cover further computer network aspects of the scenario provided and worked on in Assignment 1. For this reason, the assignment background and scenario given in Assignment 1 will also be made available in this assignment. Assignment 1 focused on the networking requirements, design and documentation of the **City1** part of the scenario. Assignment 2 will address both **City1** and **City2** networking. **The network requirements/ specifications of City1 and City2 are the same (given in the scenario).***

EXTRACTED FROM ASSIGNMENT 1:

Computer networks and network fundamentals are critical components of modern industries, connecting businesses, offices, and factories to enable communication, data sharing, and productivity. In South Africa, industries such as finance, healthcare, education, and manufacturing rely heavily on computer networks to support their operations. However, as networks become more complex and interconnected, they also become more vulnerable to cybersecurity threats.

In South African industries where data is crucial, such as finance and healthcare, networks enable quick and secure access to sensitive information. Additionally, computer networks facilitate collaboration and communication between teams, regardless of their location. It has also played a vital role in supporting business operations and connecting employees to resources such as databases, applications, and files.

In South African industries, understanding network fundamentals is necessary to optimise network performance and reduce downtime. Proper network planning, design, and implementation can improve efficiency, increase productivity, and reduce network maintenance costs. However, as networks become more ubiquitous and interconnected, network security becomes a crucial concern. Cybercriminals are increasingly targeting computer networks to steal sensitive data or disrupt business operations. Cyberattacks can result in significant financial losses, reputational damage, and loss of customer trust. Therefore, implementing robust network security measures is critical to protect against data breaches and network attacks.

To enhance network security, industries should implement a multi-layered approach that includes both preventive and reactive measures.

Preventive measures might include firewalls, intrusion detection systems, and two-factor authentication, while reactive measures might include data backup and disaster recovery plans. Educating employees on safe browsing habits, password management, and social engineering tactics can further enhance network security and minimise the risk of human error.

Computer networks have become an essential part of our lives, and their security is critical to protect against cyberattacks. South Africa, like many other countries, faces cybersecurity threats, and it is important to understand computer network fundamentals and network security. By implementing proper security measures, organisations can protect their networks from unauthorised access, theft, damage, or misuse.

ASSIGNMENT SCENARIO (EXTRACTED FROM ASSIGNMENT 1)

You have been appointed to plan and design a mega “**Library & Multimedia Technologies Centre**” network project. The network is intended to be rolled out by the government department in various municipalities and locations.

For the purpose of testing and initiating the design and its implementation, the project will be limited to **two** (2) locations: one in your local city and the other in a different city. You are required to identify the names of the two cities for reference and naming specifications. In this scenario, the cities will be referred to as **City1** and **City2**. Assignment 1 is mainly focused on City1 networking requirements, installations, and design.

The network in each city will include the following three (3) departments:

1. The Library

- a. The library must allow the borrowing and returning of print copies and e-books via the library help desk, assisted by Librarians (at least four (4) library help desks).
- b. The library must have sufficient printing, scanning, and copier devices. These devices are networked and operate via a server.
- c. The library must have 10 workstations that can be used for research (searching databases and surfing the Internet).
- d. The library must have two (2) workstations that can be used to search the availability of materials from the library. These workstations must be restricted to only allow the library system to run on them.

2. 1 x Multimedia and Technologies Facility

- a. The multimedia must have one (1) presentation boardroom.
- b. The multimedia must have three (3) cinema rooms equipped with at least one network projector.

3. 1 x Connection Lab

- a. The lab will allow users to use the devices in the lab and/ or bring their own devices.
- b. Devices include electronic devices such as smartphones, tablets and laptops.
- c. The lab must be equipped with its own workstations (20 workstations).
- d. The lab must be configured and made ready to connect up to 20 electronic devices (for users who want to bring their own devices).
- e. The lab is divided into three sections (these may be rooms):
 - i. On-site workstation section (as mentioned in (3.c) above).
 - ii. Bring-Your-Own-Device section [BYOD] (as mentioned in (3.d) above).
 - iii. Electronic Charging Spaces are equipped with UPS and power stations to allow the charging of devices of different kinds.

Mandatory: Server Room:

1. The centre must be equipped with a secure server room. Security to access the server room must be hardened.
2. The centre must also be equipped with (separate from a server room) an IT Support Office that can cater to at least three (3) IT/ Network Administrators to support the operations of the centre.

Mandatory: Use and Centre Access – Entrance/Exit

- Security must be hardened.
- Only registered members can access the Multimedia Facility.
- The lending of any material is via a system for only active registered “in good standing” members.
- The centre must be equipped with CCTV monitoring cameras in addition to assistance and security guards.

Assumptions:

- *You are required to identify and make use of your own/ preferred software and applications, where applicable.*
- *The scenario does not prescribe floor plans of any departments, server rooms, or IT support offices. You will be required to plan and design your own floor plans and partition these according to the size of the library. The book section/ space must be indicated, although you are not required to show aisle and shelving. Your floor plans must show the network architecture required.*

ANSWER ALL QUESTIONS BASED ON THE SCENARIO GIVEN ABOVE. A MARK OF ZERO (0) WILL BE AWARDED IF THE SCENARIO IS NOT APPLIED OR IGNORED. OWN ASSUMPTIONS FOR THE NETWORK SOLUTION COMPLETENESS ARE WELCOMED.

Assignment 2 Activities**(Marks: 100)**

Design a **network solution**. You are expected to make your own assumptions (which must be documented) regarding the details of the network specifications not provided in the scenario.

- Complete the activities:

Summary of Activities:

ACTIVITY	WEIGHING OF MARKS
Activity 1 – Network Architecture (Learning Unit 4)	10 marks
Activity 2 – Network Segmentation and Subnetting (Learning Unit 4)	20 marks
Activity 3 – Wide Area Networks (WANs) (Learning Unit 5)	10 marks
Activity 4 – Wireless Transmissions (Learning Unit 5)	10 marks
Activity 5 – Network Documentation (Learning Unit 2) and 1, 3, 4, and 5	50 marks
TOTAL	100 marks

ACTIVITY 1	<u>NETWORK ARCHITECTURE</u> In this activity, you are required to use the scenario and prescribed material and search the Internet to compile the solution to the following question. Outline (<i>not exceeding 1 page</i>) the details of the network architecture for the Library & Multimedia Technologies Centre network. The network architecture may include one or more of the network architecture concepts: the physical architecture, the virtual architecture, and/ or cloud architecture. Information that you have already gathered in Assignment 1 may be used.	(10)
	Marking Rubric:	

ACTIVITY 1: MARKING RUBRIC					
Criteria	Excellent/ Outstanding (4 marks)	Proficient/ Good (3 marks)	Satisfactory (2 marks)	Needs Improvement (1 mark)	Mark Allocation
Network Architecture	Clear explanation of the network's architecture, including devices, topology, and cabling.	Provides a good description of the network architecture but lacks detail in some areas.	Limited description of network architecture. Key elements are missing.	No or inaccurate description of the network architecture.	4
Application of theory	Excellent/ Outstanding (4 marks)	Proficient/ Good (3 marks)	Satisfactory (2 marks)	Needs Improvement (1 mark)	4
	Outstanding application of the theory. The network solution excellently addresses network specifications in the given scenario. Outstanding use of prescribed material, own research, and own assumption (where applicable).	Good application of the theory, but it lacks some details or scenario/ real-world application. Proficient use of prescribed material, own research, and own assumptions (where applicable).	Some application of the theory. Satisfactory use of prescribed material, own research, and own assumptions (where applicable).	No or incorrect application of the theory. Inadequate use of prescribed material, own research, and own assumptions (where applicable).	
Structure, Clarity, and Presentation	Excellent/ Outstanding (2 marks)	Proficient/ Good (1 mark)	Satisfactory (0 marks)	Needs Improvement (0 marks)	2
	Well-organised, logical flow of ideas.	Mostly well-organised with minor errors in the flow of ideas.	Somewhat disorganised, with several partially adequate flows of ideas.	Disorganised with the most inadequate flow of ideas.	
TOTAL					10

ACTIVITY 2	<u>NETWORK SEGMENTATION</u>	(20)
	In this activity, you will also refer to Assignment 1, Activity 2 (network addressing) to complete this activity.	
	Based on Activity 1 above (and other relevant information and assumptions), outline how the network will be segmented, including:	
	a. Broader network segmentation.	
	b. Subnet masks.	
	c. Subnets and IP information.	
	d. Implementation of subnets on a network.	
	Marking Rubric:	

ACTIVITY 2: MARKING RUBRIC						
Criteria		Description				Marks
Broader Network Segmentation	Clearly explains network segmentation concepts and explains how the concepts apply to the Library & Multimedia Technologies Centre.					3
	Outstanding (3 Marks)	Adequate (2 Marks)	Partially Adequate (1 Mark)	Inadequate (0 Mark)		
	Clearly explains logical and physical segmentation and network separation.	Good explanation with some minor gaps in technical details or real-world application.	General understanding but lacks relevance to the scenario.	Minimal or incorrect explanation of network segmentation.		
Subnet Masks	Provides an explanation of subnet masks, their role in segmentation, and how they improve network efficiency.					4
	Adequate (4 Marks)		Partially Adequate (3 – 2 Marks)		Inadequate (1 Mark)	
	Adequately explains subnet masks, their function, how they work with IP addressing, and their role in segmenting networks efficiently. Selected subnet mask(s) are given in the explanation.		Partially adequate understanding but lacks relevance to the scenario.		Incorrect or missing explanation of subnet mask(s).	
Subnets and IP information	Demonstrates the ability to determine subnets based on IP addressing information selected for the Library & Multimedia Technologies Centre.					10
	Adequate (8 – 10 Marks)		Partially Adequate (5 – 7 Marks)		Inadequate (1 – 4 Marks)	
	Shows adequate subnet and IP information, including IP address ranges, broadcast addresses, and the number of available hosts.		Demonstrates partial understanding but contains multiple errors. Shows partially adequate subnet and IP information, including IP address ranges, broadcast addresses, and the number of available hosts.		Incorrect calculations or a missing explanation of subnetting. Shows inadequate subnet and IP information, including IP address ranges, broadcast addresses, and the number of available hosts.	
Implementation of Subnets on a Network	Discusses practical implementation, including IP address assignment using best practices and techniques.					3
	Adequate (3 Marks)		Partially Adequate (2 Marks)		Inadequate (1 Mark)	
	Adequately discusses how to apply subnets in a real network, in this case, the proposed network solution for the Library & Multimedia Technologies Centre.		Covers the implementation but lacks practical application. Partially discusses how to apply subnets in a real network, in this case, the Library & Multimedia Technologies Centre.		Basic mention of subnet implementation with little detail. Inadequately discusses how to apply subnets in a real network, in this case, the proposed network solution for the Library & Multimedia Technologies Centre.	
TOTAL						20

ACTIVITY 3	<u>WIDE AREA NETWORKS (WANs)</u>	(10)
	Consider the requirements of the Library & Multimedia Technologies Centre network (City1 and City2), and outline (<i>not exceeding 1 page</i>) how the WANs' infrastructure and connectivity will be configured and implemented. Ensure that the outline addresses or covers the following WAN concepts: a. Endpoint equipment. b. WAN connectivity.	
	Marking Rubric:	

ACTIVITY 3: MARKING RUBRIC					
Criteria		Description			Marks
End Point Equipment	Explains the various end-point devices involved in the WAN infrastructure suitable for the Library & Multimedia Technologies Centre network and their role in connectivity (e.g., routers, switches, modems).				4
	Outstanding (4 Marks)	Adequate (3 Marks)	Partially Adequate (2 Marks)	Inadequate (0 Mark)	
	Provides a clear explanation of the various end-point equipment needed for WAN connectivity and their roles in the Library & Multimedia Technologies Centre network.	Discusses some end-point equipment and misses some key equipment required for the Library & Multimedia Technologies Centre network.	Mentions end-point equipment but lacks detail on their roles in the Library & Multimedia Technologies Centre network.	No discussion of end-point equipment.	
WAN Connectivity	Discusses the WAN connectivity methods suitable for the Library & Multimedia Technologies Centre network as well as their application in the scenario.				4
	Outstanding (4 Marks)	Adequate (3 Marks)	Partially Adequate (2 Marks)	Inadequate (1 Mark)	
	Provides a comprehensive explanation of different WAN connectivity options suitable for the Library & Multimedia Technologies Centre network with clear applications to the scenario.	Good explanation, but lacks important detail on WAN connectivity options suitable for the Library & Multimedia Technologies Centre network.	Describes basic WAN connectivity methods but lacks clarity in some areas.	Limited or no discussion on WAN connectivity methods.	
Organisation, Structure, and Clarity, Technical Accuracy, and Relevance to the given scenario	Logical flow of ideas. Demonstrates understanding of technical concepts related to WAN configuration and implementation. Includes specific examples suitable for the Library & Multimedia Technologies Centre network.				2
	Adequate (2 Marks)	Partially Adequate (1 Mark)		Inadequate (0 Mark)	
	Well-organised response with clear explanation of concepts, logical flow, and effective use of material. Demonstrates adequate technical understanding and detailed information throughout.	Organised response but may lack clarity in some areas. Technically correct but lacks detail in some areas or has minor inaccuracies.		The response is poorly organised and difficult to follow. Some inaccuracies or vague explanations, lacking technical detail.	
TOTAL					10

ACTIVITY 4	<u>WIRELESS TRANSMISSIONS</u>	(10)
	Consider the requirements of the Library & Multimedia Technologies Centre network (City1 and City2), and outline (<i>not exceeding 3 paragraphs</i>) how the wireless network infrastructure and connectivity will (1) involve, (2) configured, and (3) implemented. Ensure that the outline addresses or covers the wireless transmission concepts: topics: a. Wireless topology. b. Implementing a Wi-Fi network.	
	Marking Rubric:	

ACTIVITY 4: MARKING RUBRIC				
Criteria	Description			Marks
Wireless Topology	Describes the wireless network topology that could be implemented in this setting (the scenario).			4
	Adequate (4 Marks)	Partially Adequate (2 – 3 Marks)	Inadequate (1 Mark)	
	Provides a clear description of the wireless topology with proper application to the Library & Multimedia Technologies Centre's needs, including coverage and AP placement.	Describes the wireless topology with some detail, but it may miss key points related to network layout or coverage considerations.	General description of wireless topology with limited application to the specific scenario.	
Implementing a Wi-Fi Network	Discusses steps to implement a Wi-Fi network, including the selection of hardware and configuration settings.			4
	Adequate (4 Marks)	Partially Adequate (2 – 3 Marks)	Inadequate (1 Mark)	
	Adequate explanation of the steps involved in implementing the Wi-Fi network, including hardware selection and configuration.	Good explanation, but may lack detail in some areas.	Describes basic implementation steps but lacks key areas.	
Organisation, Structure, and Clarity. Technical Accuracy and Depth	Logical flow of ideas. Demonstrates understanding of technical concepts related to wireless network configuration and implementation. Includes specific examples where applicable.			2
	Adequate (2 Marks)	Partially Adequate (1 Mark)	Inadequate (0 Mark)	
	Well-organised response with clear and logical flow, and effective use of material. Demonstrates adequate technical understanding throughout.	Organised response but may lack clarity in some areas. Demonstrates partially adequate technical understanding.	The response is poorly organised and difficult to follow. Some inaccuracies or vague explanations, lacking technical understanding.	
TOTAL				10

ACTIVITY 5 NETWORK DOCUMENTATION

In this activity, you are required to use the prescribed material and search the internet to compile a list of the required information in the following questions.

You will also refer to Assignment 1, Activity 4 (network documentation) to complete this activity.

Consider the definitions and related content in the prescribed material and then complete the activity.

A WAN (Wide Area Network) diagram is a specific type of network diagram that focuses on a network spanning a large geographic area, while a network diagram is a broader term encompassing any visual representation of a network, including both physical and logical layouts, and a logical diagram focuses on the abstract data flow and communication protocols, rather than the physical infrastructure.

Here is a more detailed breakdown:

1. Network Diagram (General):

- **Definition:** A network diagram is a visual representation of a computer or telecommunications network, showing the components (devices, servers, etc.) and their connections.
- **Purpose:** Helps network administrators and engineers understand the network's structure, troubleshoot issues and plan for future upgrades or changes.
- **Types:** Can be either physical or logical.

2. Physical Network Diagram:

- **Focus:**
Depicts the physical layout of the network, including cables, ports, racks, and the physical location of devices.
- **Example:**
Shows where routers, switches, and servers are located, and how they are connected via cables or wireless connections.

(50)

- **Use Cases:**

Useful for physical infrastructure planning, cable management, and troubleshooting physical connectivity issues.

3. Logical Network Diagram:

- **Focus:**

Illustrates how data flows through the network, focusing on the network topology, protocols, and communication paths, rather than the physical infrastructure.

- **Example:**

Shows subnets, VLANs, routing protocols, and how different devices communicate with each other.

- **Use Cases:**

Useful for network design, troubleshooting network connectivity issues, and understanding how data flows through the network.

4. WAN Diagram (Specific Type of Network Diagram):

- **Definition:**

A WAN diagram is a type of network diagram that focuses on a network that spans a wide geographic area, connecting multiple LANs (Local Area Networks) or other networks.

- **Focus:**

Shows how different locations or offices are connected, and how data is transmitted between them.

- **Example:**

It might show a company's network connecting multiple branch offices, or a network connecting different cities or countries.

- **Use Cases:**

Useful for designing and managing WAN infrastructure, troubleshooting connectivity issues between different locations, and optimising network performance.

Useful resources (not compulsory):

Website

- https://www.conceptdraw.com/solution-park/CN_TOOL_COMPNETDIAGRAMS

Video:

- Cisco Packet Tracer Project 2022 available at:
<https://www.youtube.com/watch?v=e1cD2Klme-E>

NB: The video clip is long. Consider your Internet access (wi-fi or data) before accessing and watching the video.

Use the Packet Tracer to complete the tasks below, ensuring that all requirements in **NOTE (i)** below are adequately addressed:

NOTE (i):

You are expected to configure the core devices (network devices) and end devices (host devices) to provide end-to-end connectivity and access internal servers and the external server.

1. Use segmentation principles to assign separate IP networks.
2. The switches should be configured with appropriate VLANs and security settings.
3. RIPv2 will be used to provide routing for the routers in the internal network and static routing for the external server.
4. The devices will be expected to acquire dynamic IP addressing from a router-based DHCP server.

TASK

1. Plan, design and prototype the network topology for the entire (City1 and City2) **Library & Multimedia Technologies Center**. *Assume that City1 is the main network and City2 is a branch network.* Configure the Packet Tracer network with appropriate settings to achieve the connectivity and functionalities specified in the requirements. Follow the following sub-sections:

	a. Network and host devices. i. Selection and placement of router devices. ii. Selection and placement of switch devices. iii. Selection and placement of host devices. b. Connections. i. Configuration of router connections. ii. Configuration of switch connections. iii. Configuration of host device connections. c. Show network segmentation. d. Show VLSM and VLANs. e. Show subnetting using the correct CIDR notation. f. Configure the network nodes and protocols. i. Configuration of router devices (core layer). ii. Configuration of the layer 2 switches (distribution layer). iii. Configuration of the layer 3 switches (access layer). g. Show the WAN design and connectivity. h. Show labelling of all network nodes, boundaries, connections, VLANs, segmentation and subnetting.	
2.	Save the packet tracer file (.pka or .pkt) in a directory and submit the file with your assignment document as evidence of your diagram.	
3.	Produce a report and include it in your assignment document under Activity 5. The report must include: a. Take a screenshot of the topology diagram created in the Packet Tracer. ○ The diagram must be clear and show all the details and labelling as required. ○ A short description (approximately 1 paragraph) of what the diagram is about.	

Requirement		Mark Allocation
1 (a) Selection and placement of network and host devices.		11
• Adequately selected and placed router devices (core).	3	
• Adequately selected and placed switch devices (core).	3	
• Adequately selected and placed host devices (end).	3	
• Adequately labelled the devices using appropriate naming conventions.	3	
1 (b) Connections (interface) configuration and 1 (f) Configure the network nodes.		15
• Adequately placed and configured the router connections (core to distribution layer): City2	3	
• Adequately placed and configured the switch connections (distribution to access layer): City2	3	
• Adequately shown the connection between the main and branch networks.	3	
• Adequately shown host device and host device connections.	3	
• Adequately labelled host devices using appropriate naming conventions.	2	
1 (c) Network segmentation		4
• Topology diagram adequately shows network segmentation that would work for the given scenario.	2	
• Each network segment is adequately labelled using appropriate naming conventions.	2	
1 (d) VLAN		5
• Topology diagram adequately shows VLANs that would work for the given scenario.	3	
• VLAN principles are adequately applied.	2	
1 (e) Show subnetting (CIDR notation) in the topology diagram		5
• Adequately shown subnetting based on segmentation and interfaces.	3	
• Adequately selected and applied subnetting principles for each subnet based on the scenario.	2	

1 (g) WAN		5
• The topology diagram shows the placement and connection of the WAN adequately.	3	
• Adequately labelled the connections/ interfaces using appropriate naming conventions.	2	
3 Packet Trace File		5
• Packet Tracer file is submitted.		
TOTAL		50
<u>3 Mark Allocation</u> 3 marks – Adequate; 2 marks – Partially adequate; 1 mark – Inadequate; 0 mark – Not included or not acceptable.	<u>2 Mark Allocation</u> 2 – Adequate; 1 – Inadequate; 0 mark – Not included or not acceptable.	

[TOTAL MARKS: 100]